

Infrared radiation (RP)

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

QUESTION		ANSWER	✓
1	Describe a Leslie cube.	A hollow metal cube with faces of different colours and textures, filled with hot water.	☐
2	Infrared emission from surfaces: name the independent variable.	The type of surface	☐
3	Infrared emission from surfaces: name the dependent variable.	The amount of infrared radiation emitted	☐
4	Infrared emission from surfaces: name the control variables.	Temperature, area and material of the surface, and its distance from the detector	☐
5	Describe how the infrared radiation emitted by a surface is measured.	With an infrared detector at a fixed distance.	☐
6	State how the reading on an infrared thermometer depends on the radiation it detects.	The more radiation detected, the higher the reading.	☐
7	All faces of a Leslie cube are at the same temperature. Explain why the infrared readings differ.	Different surfaces emit at different rates: matt black emits most, shiny silver least.	☐
8	State the type of surface that is the best emitter of infrared radiation.	Matt black	☐
9	State the type of surface that is the worst emitter of infrared radiation.	Shiny silver	☐
10	State the type of surface that is the best absorber of infrared radiation.	Matt black	☐